Introduction to Large Language Models

A short beginner guide - sample document for Ask Your PDF

1. What is a Large Language Model?

A Large Language Model (LLM) is a deep learning system trained on huge amounts of text such as books, websites, articles and code. It learns statistical patterns in language so that it can read a prompt and generate a useful reply. Famous examples include ChatGPT, Claude and Gemini.

The word **Large** refers to the enormous size of the training data and the billions of parameters inside the model. The word **Language** means that it works with human languages like English or Hindi and with programming languages like Python. The word **Model** means it is a neural network that learns from data.

2. How does an LLM generate text?

An LLM does not search Google for every answer and it does not think like a human. It predicts the next token again and again. The process has these steps: the text is split into tokens, tokens are converted into numbers, the numbers are processed by Transformer layers, and finally the model picks the most likely next token and repeats the loop until the answer is complete.

A token is a small piece of text. It can be a whole word, part of a word or a punctuation mark. For example the sentence 'What is Python?' can become four tokens: What, is, Python and the question mark.

3. Embeddings

An embedding is a list of numbers (a vector) that represents the meaning of a piece of text. Sentences with similar meaning get vectors that are close to each other. For example, 'I enjoy building AI projects' and 'I want to become an AI engineer' will be much closer than 'My favorite food is pizza'.

Embedding models such as all-MiniLM-L6-v2 from Hugging Face turn every sentence into a vector of 384 numbers. Embeddings are the foundation of semantic search and of RAG applications.

4. Vector databases

A vector database stores embeddings and can quickly find the vectors that are most similar to a query. FAISS, created by Meta, is a popular open source library for this job. Chroma and Pinecone are other well known options. When you ask a question, the question is also converted into a vector and the database returns the closest chunks of text.

5. Chunking

Long documents are split into smaller pieces called chunks before they are embedded. Smaller chunks make search more precise, and an overlap between neighbouring chunks makes sure that no sentence loses its meaning because it was cut in half. A common choice is a chunk size of about 1000 characters with an overlap of 100 to 200 characters.

6. Retrieval-Augmented Generation (RAG)

LLMs only know what they saw during training. They do not know your private documents and they do not update their knowledge in real time. RAG solves this problem. First, the documents are split into chunks and stored as embeddings. Second, when the user asks a question, the most relevant chunks are retrieved. Third, those chunks are placed inside the prompt and the LLM writes an answer using only that information.

The biggest benefits of RAG are fewer hallucinations, answers that can be traced back to a source page, and the ability to use private data without retraining the model.

7. Prompts

A prompt is the instruction given to the model. Modern chat models use roles. The system message defines the behaviour of the assistant, the user message contains the question and the assistant message contains the previous reply. A prompt template is a reusable prompt with variables, for example 'Explain {topic} in simple terms'. A structured prompt asks the model for a fixed format such as JSON.

8. Memory

Without memory every message is treated independently. With memory the model receives the earlier messages and can answer follow-up questions. Because the history grows with every message, good applications keep only the last few messages to control cost and avoid the token limit.

9. Temperature and max tokens

Temperature controls how random the output is. A temperature of 0.0 gives predictable answers, 0.3 gives focused answers, 0.7 gives creative answers and 1.0 gives very varied answers. For question answering over documents a low temperature such as 0.2 is recommended. Max tokens sets the maximum length of the generated answer. Tokens are not exactly the same as words.

10. API model or local model?

There are two ways to use an LLM. With an API the model runs on the servers of a provider such as Groq, OpenAI or Google. You only send a prompt and receive a response, but you need an internet connection and must keep your API key private. With a local model you download the weights, for example from Hugging Face, and run the model on your own computer. This works offline and is free, but it needs enough RAM and processing power.

11. Limitations

LLMs can make mistakes and can sound confident while being wrong. This is called a hallucination. They also have a context window, which is the maximum number of tokens they can read at once. Scanned PDFs that contain only images cannot be read without OCR, so text based PDFs work best.